



Predicting municipal solid waste gasification using machine learning: A step toward sustainable regional planning

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ABSTRACT

The gasification process can treat and valorize municipal solid waste (MSW) in an environmentally and economically friendly way. Using this process, MSW can be safely disposed of and sustainably converted into bioenergy as part of regional planning. Experimental laboratory data is a key component in designing, optimizing, controlling, and scaling up MSW gasifiers. However, most researchers lack the resources and time to conduct experiments. Machine learning (ML) technology can resolve this issue by detecting patterns and hidden information in published data. Hence, the present study aims to construct an inclusive ML model to predict and understand the MSW gasification process. The objective is to establish a consistent and homogeneous database containing MSW sources under different gasification conditions, followed by an analysis of the database using statistical methods. Three ML models are used to predict the distribution of syngas, char, and tar and the quality of syngas in MSW gasification using feedstock characteristics and gasification parameters. When a gradient boost regressor is used to model the process, the prediction accuracy is highest ($R^2 > 0.926$, RMSE < 6.318 , and RRMSE < 0.304). SHAP analysis is successfully used to understand the significance and contribution of descriptors on targets in the modeling process.

1. Introduction

A steady increase in the quantity of municipal solid waste (MSW) is evident worldwide because of growing urbanization, improving living standards, and increasing economic growth [1]. This excessive increase in MSW has posed significant challenges for regional governments because of the lack of space and the high disposal costs [2]. When MSW is not properly managed, it can cause severe health and environmental problems by polluting the air, water, and soil [3]. Despite numerous

advancements in MSW management systems (prevention, reduction, separation, recycling, and energy recovery) [4], the amount of MSW treated by landfilling, dumping, and incineration is still of great concern [5]. These traditional MSW treatment methods have resulted in serious environmental problems, such as methane emissions and leachate release from landfills and carbon dioxide and dioxin emissions from incinerators [6].

Despite concerns about its impact on human health and the environment, MSW can be used in modern waste treatment systems to

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Table 1
Some ML models used in the literature to predict the gasification process.

Feedstock type(s)	Gasification type	Selected model(s)	Model input(s)	Model output(s)	Model limitation(s)	Statistical results	Ref.
Olive pits	Conventional	Linear Regression	Time, temperature, CO, CO ₂ , CH ₄ , O ₂ , and higher heating value (HHV) of syngas	H ₂ yield	-Applicable only for one type of feedstock -Ignoring the equivalence ratio as an input feature -Applicable only for predicting hydrogen yield	$R^2 \approx 0.99$, $0.008 < \text{MSE} < 2.66$, $0.007 < \text{MAE} < 0.762$	[16]
Pine cone, wood pellet	Conventional	Polynomial regression, decision tree	Feedstock ultimate analysis (excluding sulfur content), feedstock proximate analysis, equivalence ratio, feedstock flow rate, and temperature distribution	Yield (CO, CO ₂ , CH ₄ , and H ₂) and syngas HHV	-Applicable only to certain feedstock types	$0.80 < R^2 < 0.94$, $0.22 < \text{RMSE} < 1.21$, $0.04 < \text{NRMSE} < 0.19$	[17]
Pine cone, wood pellet	Conventional	Nonlinear autoregressive with exogenous neural networks	Feedstock ultimate analysis (excluding sulfur content), feedstock proximate analysis, equivalence ratio and its one- and two-step delayed, one-step delayed CO, CO ₂ , CH ₄ , H ₂ , syngas HHV, and outlet gas temperature	Yield (CO, CO ₂ , CH ₄ , and H ₂), syngas HHV, and outlet gas temperature	-Applicable only to certain feedstock types	$0.83 < R^2 < 0.87$, $0.47 < \text{RMSE} < 1.26$, $0.05 < \text{NRMSE} < 0.08$	[18]
Twenty different biomass feedstocks	Conventional	Multilayer perceptron neural network	Oxygen content, hydrogen content, carbon content, temperature, steam flow rate, and feedstock flow rate	Exergy value of syngas and H ₂ yield	-Modeling of limited outputs	$R^2 \approx 0.99$	[19]
Different biomass feedstocks	Supercritical water	Random forest regression	Carbon content, hydrogen content, oxygen content, ash content, biomass concentration, gasification temperature, reactor pressure, and residence time	H ₂ yield	-Neglecting proximate analysis -Using limited data patterns (only 95 data patterns) -Overlooking the equivalence ratio as an input -Modeling of limited outputs	$R^2 \approx 0.97$, $\text{RMSE} \approx 1.02$, $\text{MAEP} \approx 10.61$	[20]
Coal	Supercritical water	Gradient boosting regression	Feedstock ultimate analysis, feedstock proximate analysis, temperature, equivalence ratio, concentrator, and reaction time	H ₂ yield	-Applicable only to certain feedstock types -Modeling of limited outputs -Applicable only for predicting hydrogen yield	$0.96 < R^2 < 0.97$, $0.01 < \text{MSE} < 0.46$, $3.29 < \text{MRE} < 7.82$	[21]
Different biomass feedstocks	Conventional	Multilayer perceptron neural network	Carbon content, hydrogen content, sulfur content, moisture content, ash content, particle size, temperature, equivalence ratio, steam-to-biomass ratio, operation mode, catalyst (used or not used), process scale, gasifying agent, reactor type, and gasifier bed material	Yield (N ₂ , H ₂ , CO, CO ₂ , CH ₄ , and C ₂ H ₄), syngas lower heating value (LHV), and yield (syngas, char, and tar)	-Modeling using inconsistent data patterns (such as non-catalytic and catalytic gasification data without including catalyst properties)	$R^2 \approx 0.93$, $0.13 < \text{RMSE} < 0.21$	[22]
MSW	Conventional	Gradient boosting regression	Feedstock ultimate analysis, feedstock proximate analysis, feedstock constituents (cellulose, hemicellulose, and lignin), temperature, equivalence ratio, pressure, biomass-to-steam ratio, operating condition (batch and continuous), gasifying agent, reactor type, bed material, catalyst (used or not used), and process scale (pilot or lab)	Yield (N ₂ , H ₂ , CO, CO ₂ , CH ₄ , and C ₂ H ₄), syngas LHV, and yield (syngas, char, and tar)	-Modeling using inconsistent data patterns (e.g., non-catalytic and catalytic gasification data without considering catalyst properties) -Insufficient data patterns due to a high number of input variables	$0.76 < R^2 < 0.96$, $0.4 < \text{RMSE} < 10.29$	[23]

produce renewable biofuels and biochemicals [7]. Various biochemical (i.e., anaerobic digestion, fermentation, and composting) and thermochemical (i.e., pyrolysis, gasification, and hydrothermal processing) approaches have been developed and commercialized around the world to valorize MSW [8,9]. Biochemical pathways have recently attracted significant research attention in treating MSW because of their mild operating conditions in terms of temperature and pressure. MSW is a

heterogeneous mixture of wastes with different physicochemical properties, so biochemical treatment cannot handle all its components simultaneously. Moreover, some pretreatment processes are necessary to improve the digestibility of MSW components, even biological ones. These issues make the biochemical treatment of MSW a technically complicated, cost-intensive, and time-consuming option.

With little or no pretreatment, thermochemical routes can handle

most MSW components more efficiently than biochemical pathways. Among various thermochemical platforms developed for MSW valorization, gasification technology has attracted much interest because of its environmental and economic benefits [10,11]. Gasification outperforms incineration in terms of energy efficiency and exhaust emissions. The capital cost of MSW gasification plants is similar to those of conventional waste-to-energy systems [4]. Gasification has a much higher conversion rate than pyrolysis. In addition, the solid residue left in the gasification process is much lesser than in the pyrolysis process. Unlike hydrothermal systems, gasification plants do not require longer processing times and higher capital costs.

In MSW gasification, higher processing temperatures are used in the presence of gasifying agents (such as air, oxygen, steam, and CO₂). The main product of the gasification process is syngas, which comprises CO, H₂, CO₂, CH₄, and trace impurities [12]. This gaseous fuel can be upgraded using various cleaning and conditioning methods and used in gas engines, turbines, and fuel cells to produce power and in chemical and biochemical systems to synthesize liquid fuels [10,13]. Generally, gasification systems can play a key role in sustainable regional planning because they can safely dispose of MSW and efficiently convert it to clean energy carriers [14]. Waste management can be made safer and cheaper with these thermochemical systems for local municipalities and village councils. MSW gasification plants can also fully materialize the circular economy without generating secondary pollutants.

Despite the promising features of MSW gasification, it is an ill-defined phenomenon governed by several endogenous (i.e., MSW composition and its characteristics) and exogenous (gasifying agent, gasification temperature, gasifying agent-to-MSW ratio, reactor configuration, and residence time) parameters. The gasification process also involves transient heat, mass, and momentum transfer with concurrent chemical reactions in a multi-phase system. Therefore, using phenomenological approaches (e.g., computational fluid dynamics) to model such a complicated process is challenging. The experimental data obtained in the laboratory is used primarily to design, optimize, control, and scale up MSW gasification systems. Therefore, many experiments must be conducted to find the optimum input variables to maximize product yield. Nevertheless, time and resources often make it impossible for researchers to conduct numerous experimental trials. Designing and optimizing MSW gasifiers based on experimental observations does not always increase the probability of success enough.

The bottlenecks mentioned above have been overcome through machine learning (ML). ML technology can predict complex chemical reactions at a low cost and with high accuracy. Moreover, this technology can deal with massive, high-dimensional datasets with reasonable computational costs [15]. ML method applies advanced computational processes to analyze training data and learns to establish accurate models. This modeling approach allows large gasification datasets to be processed and interpreted more effectively than conventional statistical approaches. The accurate data-driven models provided by ML can then be used to establish key relationships between the independent process parameters and the dependent desired targets. By using ML, it is possible to uncover trends and patterns within gasification datasets without having to understand the underlying physicochemical mechanisms. As a result of their unique features, ML technology is being used to model the gasification of various feedstocks in several research publications. Table 1 details some ML technology available in the literature to predict the gasification process.

According to the published literature, ML can reliably model the gasification process by learning from experimental data without being explicitly programmed. Thanks to the rapid growth of data available in MSW gasification, ML can be applied at the moment to make highly accurate predictive models and understand the associations between input variables and output responses of this highly nonlinear process. Recently, Ascher et al. [23] attempted to develop a universal ML model for MSW gasification. Despite the promising results reported, the developed model suffers from several shortfalls. For example, they used

both catalytic and non-catalytic gasification data patterns in the modeling process without considering the catalyst properties. Although some feedstocks were plastic wastes, biomass constituents (cellulose, hemicellulose, and lignin) were input features. Thus, this study is devoted to constructing an inclusive ML model to predict and understand the non-catalytic MSW gasification process. A database containing various MSW sources under different gasification conditions reported in the literature is first established. Efforts are made to compile a consistent database to improve the accuracy and reliability of ML models. The database is then investigated using statistical analyses and discussed using mechanistic explanations. Three ML approaches, i.e., support vector machine (SVM), random forest regression (RFR), and gradient boost regressor (GBR), are applied to characterize the MSW gasification process as a function of biomass characteristics and gasification parameters. SHapley Additive exPlanations (SHAP) analysis is finally used based on the best-performing model to reveal and realize the importance and contribution of independent input parameters on dependent output responses in the modeling process.

2. Research method

The first step in this study was to compile the data patterns required for ML modeling. To this end, all articles published with the related subject were searched and downloaded to gather a trustable databank for modeling the MSW gasification process. The eligible papers were initially screened with a focus on experimental ones. Then, independent input features and dependent output responses were chosen based on available data. The independent input features include MSW ultimate characteristics, MSW proximate characteristics, and gasification operating parameters, while the dependent output responses are syngas yield, syngas composition, syngas LHV, tar yield, and char yield. The required data were extracted from the papers (tables and graphs), and a trustable databank was created. The collected data patterns were carefully checked to remove outliers. A careful check was made to remove outliers from the collected data patterns. The missing input data were imputed before ML modeling. The gathered data were also analyzed to improve understanding of the biomass gasification process. The prepared dataset was subjected to three ML models (i.e., SVM, RFR, and GBR) in order to model the MSW gasification process as a function of biomass characteristics and gasification parameters. Three statistical criteria were used to find the best ML model. As a final step, the SHAP method was applied for feature importance analysis to determine the effect of independent input variables on the dependent output responses.

2.1. Data compilation

The Scopus database was chosen to prepare the dataset required in ML modeling. For data compilation, the keywords “Municipal solid waste” and “Gasification” were used in searching the database within the article title, abstract, and keywords. Initially, 904 articles were found. After a more detailed screening, 44 articles were eligible for data extraction. The MSW used in the published papers included food wastes, cardboard, plastic, textile, rubber, household wastes, papers, and biomass wastes. The eligible articles contained the necessary input data. The input features include MSW ultimate characteristics (C, H, N, S, and O content), MSW proximate characteristics (ash content, volatile matter, and fixed carbon), and gasification parameters (reaction temperature and equivalence ratio). The output variables are the output variables: syngas yield, syngas composition, syngas LHV, tar yield, and char yield. The syngas composition includes the mole distribution of H₂, CO, CO₂, and CH₄. Data from the eligible papers were collected from their tables and graphs. For extracting data from graphs in the eligible papers, “Plot digitizer” software was used. Overall, 317 data patterns were obtained from eligible papers (Supplementary Excel File). It should be noted that dry-basis proximate data was used in the modeling process. Therefore,

the reported wet-basis proximate data were converted to dry-basis ones to harmonize the compiled dataset (red-inlined cells in the Supplementary Excel file).

Much data must be collected to develop an effective ML model. The amount of data required for ML modeling is difficult to estimate. The number of data patterns required correlates strongly with the complexity of the problem and the algorithm used to solve it [24]. When developing ML models, the number of required data patterns is recommended to be about 5–10 times the number of independent input parameters [25]. ML modeling seems feasible for all the output responses based on the collected data patterns. To be more specific, the minimum sample size of 62 data patterns for char and tar yield suffices to develop ML without overfitting. Before starting the modeling step, the missing input features should be approximated or reconstructed using an appropriate method (green-inlined cells in the Supplementary Excel file). Missing data of input descriptors (mostly some proximate analysis data) were imputed using the k-nearest neighbor (KNN) method. This non-parametric or lazy ML technique works based on proposing similar data to the same class based on the higher probability. The KNN technique does not need any definitive training process, and the major class based on the KNN value is considered in predicting missing data [26]. In the current study, the number of neighborhoods was equal to five. The collected data patterns were normalized between 0 and 1 before data imputation to obtain better data reconstruction.

2.2. Modeling

This study used three supervised ML models, including SVM, FRF, and GBR, to model the MSW gasification process. SVM is a computational learning technique based on structural risk minimization. Structural risk minimization is rooted in statistical learning theory (Vapnik–Chervonenkis theory) [27]. The SVM method can predict dependent parameters from finite samples by building a decision function by defining an optimal hyperplane that classifies and separate different observation based on their features. The hyperplane found is then applied to unseen data for labeling and classifying. It should be noticed that the features used to define the hyperplane are often derivative data resulting from the interpolation of raw data during the feature selection. The hyperplane defined by maximizing class margins can be linear or nonlinear [28]. The SVM machine can be used in regression problems to find relationships between dependent output parameters (y) and a set of independent variables ($x_1, x_2, \dots, x_n$). In the regression problem, the parameter ε is the margin to measure the discrepancy between the real and estimated values. Then, the summation of ε_i is minimized through optimization methods (e.g., least-square) to find the optimum function. Despite good generalization and training of the SVM model, finding an optimum kernel function and its hyperparameters is difficult, particularly for large datasets [29].

The RFR model is another supervised ML technique for solving regression problems. Random forests are based on the ensemble method and are built from a combination of tree predictors. Each tree is trained through a bootstrap sampled from the training dataset [30]. To solve regression problems through the random forest, the problem starts from the root of a tree and then divides it into two branches based on variable outcomes. The process is continued until reaching the leaf node and achieving final results. Once the tree is shaped, a testing data point can navigate the previously built particular path onto the particular leaf according to training. In this step, the associated value for output can be examined using statistical parameters for validating the model [31]. Several important hyperparameters, including $n_{\text{estimators}}$ (number of decision trees), criterion (determination of loss function to regulate model outcomes), max_depth (extreme possible depth of each tree), max_features (maximum number of features in splitting process), bootstrap (if model tracks bootstrapping principles), max_samples (biggest size of each sample for each tree in bootstrapping) can be tuned to optimize the RFR results [32].

The GBR model belongs to the ensemble method category. In this ML technique, several weak learners are created and combined to build a strong learner [33]. The training process in the GBR model is comparable with the random forest technique, but some differences make GBR unique. In the random forest method, each tree is trained independently. However, the GBR model is a greedy system in which all the trees are trained once at a time, and then the final results are based on their prediction sum up. Therefore, by fitting each tree step by step according to the residuals of the previous tree, the newly built tree can focus on the previous errors and correct the error value as much as possible. The first fitting decision tree is defined as: $F_1(x) = y$, and the next one is fitted by previous tree residual ($h_1(x)$) through the next fitted function $F_2(x) = F_1(x) + h_1(x)$. This process is continued until reaching the best fitting model ($F_n(x)$) [34]. The GBR model is a strong technique to fit nonlinear problems and shows great performance when faced with missing values, outliers, and high cardinality categorical in the training dataset. Because of these features, the GBR model shows great prediction performance and robust output [35].

All the input variables and output parameters were normalized between 0 and 1 before being used in ML modeling. The training phase used $\approx 80\%$ of the collected data, while the testing used the remaining $\approx 20\%$. In addition, the k-fold resampling method technique was applied to avoid overfitting and improve the prediction capability of the applied ML approaches [36]. Therefore, the training dataset was separated into five partitions. Four of the five folds were used for training, while the fifth was used for validation. The validated models were finally introduced to the testing phase with the remaining $\approx 20\%$ of the collected dataset. The capability of each developed model in the training and testing phases was evaluated through three statistical criteria (i.e., coefficient of determination (R^2), root-mean-square-error (RMSE), and relative-root-mean-square-error (RRMSE)).

2.3. SHAP feature analysis

Because of the black-box character of most ML approaches, it is not easy to interpret model features and understand how they work. Therefore, using a feature importance analysis is crucial to comprehend the importance and contribution of each independent input feature on output parameters in ML modeling. The SHAP feature importance analysis was applied to explore the importance and contribution of each independent input variable on output targets. The SHAP technique, introduced by Scott Lundberg and Su-In Lee [37], works based on removing a specific variable and tracking how estimation can change and influence the modeling process. The SHAP value can compute both the magnitude and direction (positive or negative) of the effects of input variables on dependent outputs [38].

The SHAP analysis has been successfully applied to several biomass thermochemical conversion studies, such as wastewater sludge pyrolysis [26], biomass microwave pyrolysis process [39], hydrothermal processing [40], and biomass and plastic co-pyrolysis process [41]. Shahbeik et al. [24] examined the effect of feedstock properties and reaction conditions on the quantity and quality of bio-oil, syngas, and biochar produced from wastewater sludge pyrolysis using the SHAP method. Bio-oil yield was primarily affected by ash content, fixed carbon content, operating temperature, and volatile matter content. Yang et al. [39] applied the SAHP analysis to examine the effects of biomass properties and operating conditions on the quantity and quality of the products generated in biomass microwave pyrolysis. Operating temperature, microwave power, and reaction time greatly affected product distribution and quality.

Leng et al. [40] used the SHAP method to understand the effects of biomass properties and operating conditions on the pH, total nitrogen, total organic carbon, and total phosphorus of the aqueous phase generated in hydrothermal treatment. The biomass nitrogen content, solid content, and processing temperature most influenced the pH, total nitrogen, and total phosphorus of the aqueous phase, while total organic

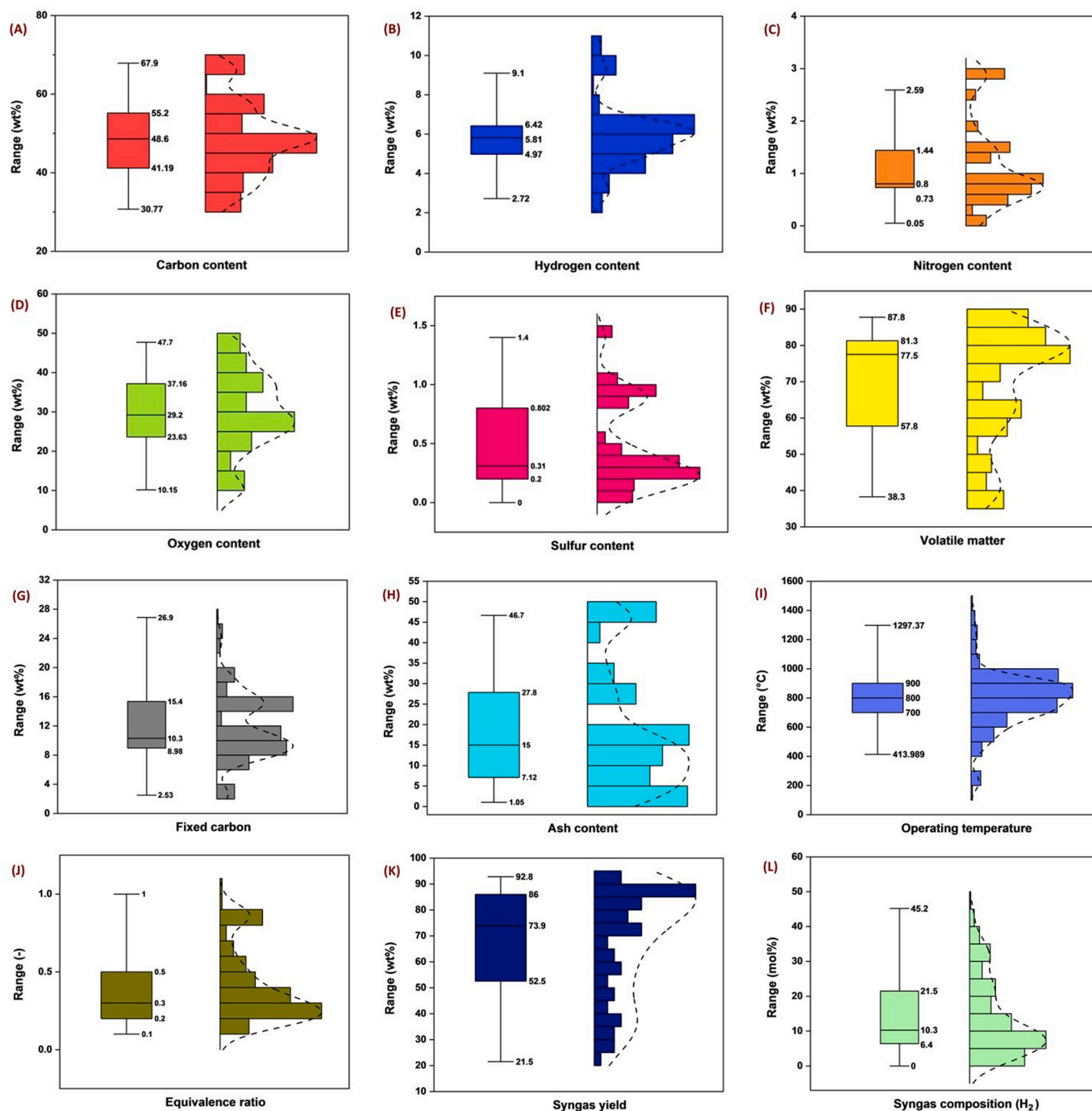


Fig. 1. Statistical analysis of modeling data. (A) MSW carbon content; (B) MSW hydrogen content; (C) MSW nitrogen content; (D) MSW oxygen content; (E) MSW sulfur content; (F) MSW volatile matter; (G) MSW fixed carbon; (H) MSW ash content; (I) operating temperature; (J) Equivalent ratio; (K) syngas yield; (L) syngas composition (H₂); (M) syngas composition (CO₂); (N) syngas composition (CH₄); (O) syngas composition (CO), (P) syngas lower heating value (LHV), (Q) char yield; (R) tar yield.

carbon was influenced by reaction time, lipid content, and temperature. Prasertpong et al. [41] evaluated the effects of 26 input features on the pyrolysis oil yield of biomass and plastic waste co-pyrolysis using the SHAP method. The most important factors affecting pyrolysis oil yield were reaction temperature and biomass-to-plastic ratio. Nevertheless, biomass and plastic characteristics contributed over 60% to the pyrolysis oil yield.

3. Results and discussion

3.1. Statistical analysis of modeling data

Modeling data was subjected to statistical evaluation to find out the association between parameters of interest in the MSW gasification process. The boxplot analysis (box range, whisker, and median value) and data distribution for the input/output parameters are depicted in Fig. 1. The carbon content of MSW ranges from 30.77 wt% to 67.90 wt%,

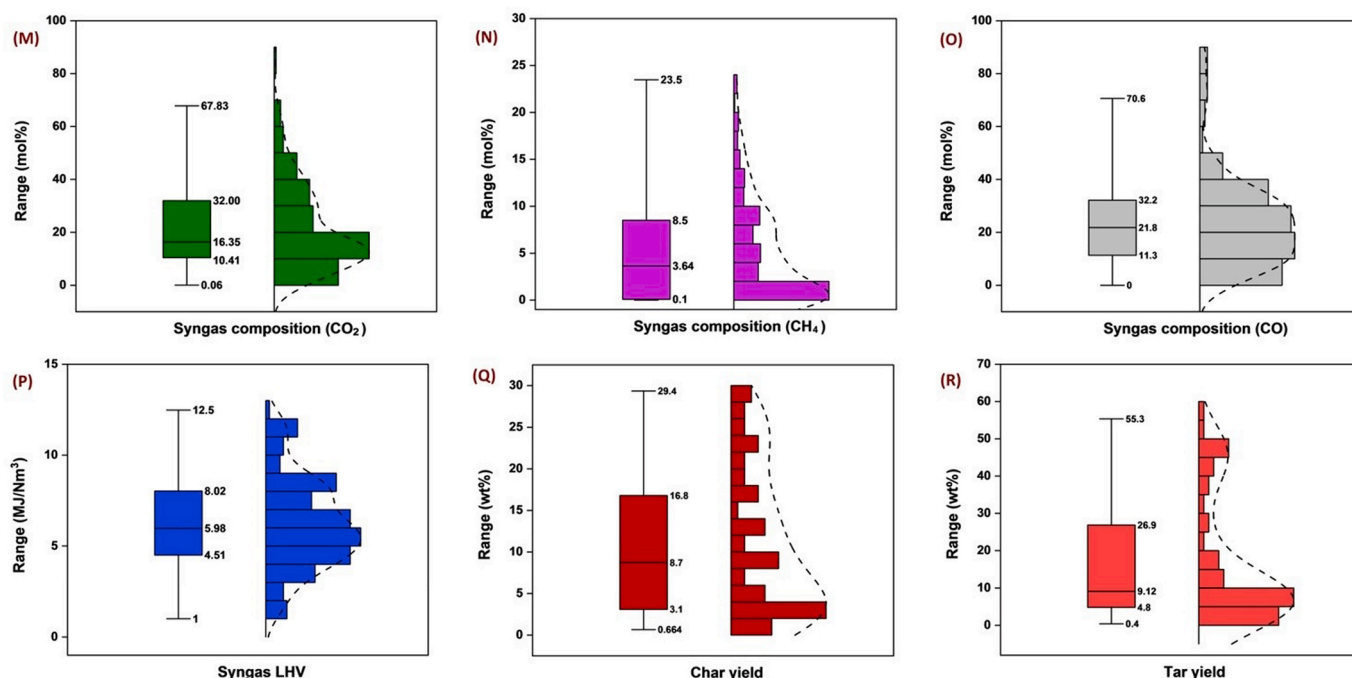


Fig. 1. (continued).

with a median value of 48.60 wt%. The high carbon content of MSW indicates its high potential as a feedstock for biofuel production. The hydrogen, nitrogen, and oxygen contents of MSW are in the range of

2.72–10.70 wt%, 0.05–2.90 wt%, and 10.15–47.70 wt%, respectively. The volatile matter of MSW varies from 38.29 wt% to 87.78 wt%, while the fixed carbon and ash content are distributed from 2.53 wt% to

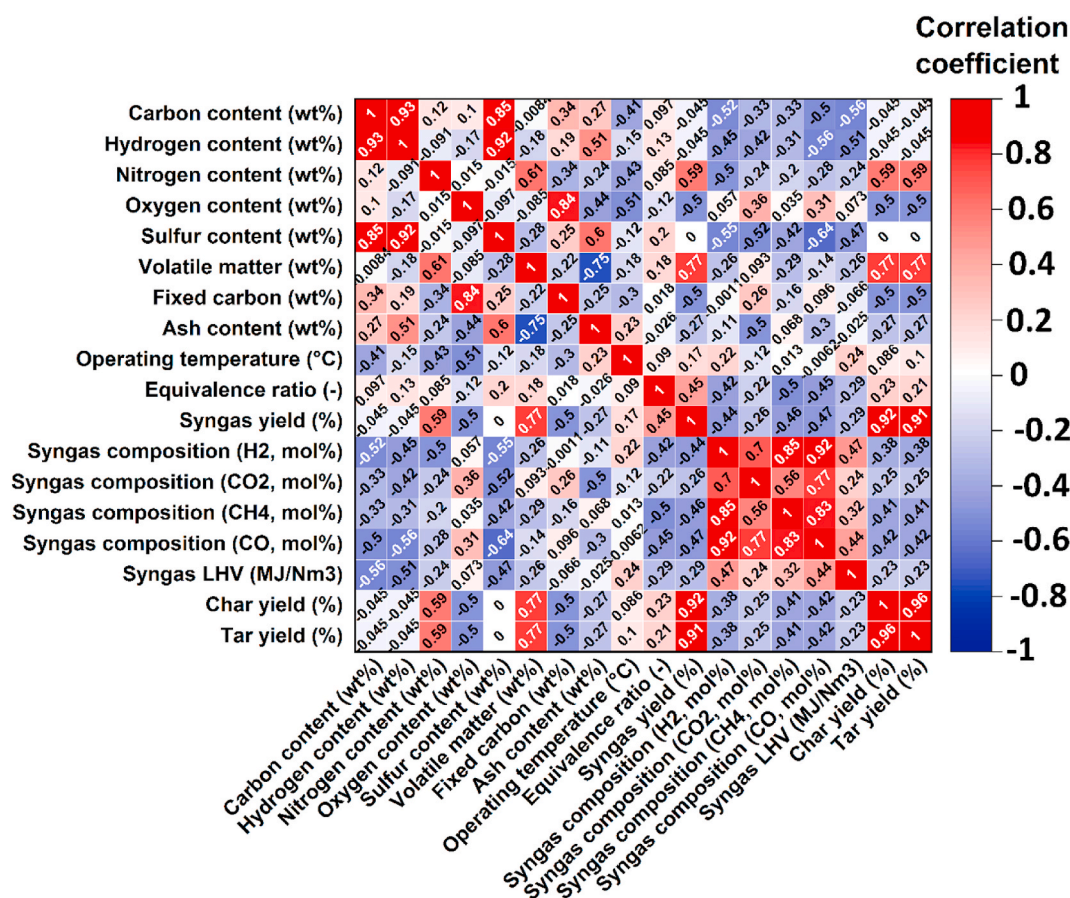


Fig. 2. Spearman correlation matrix between any two variables in modeling data.

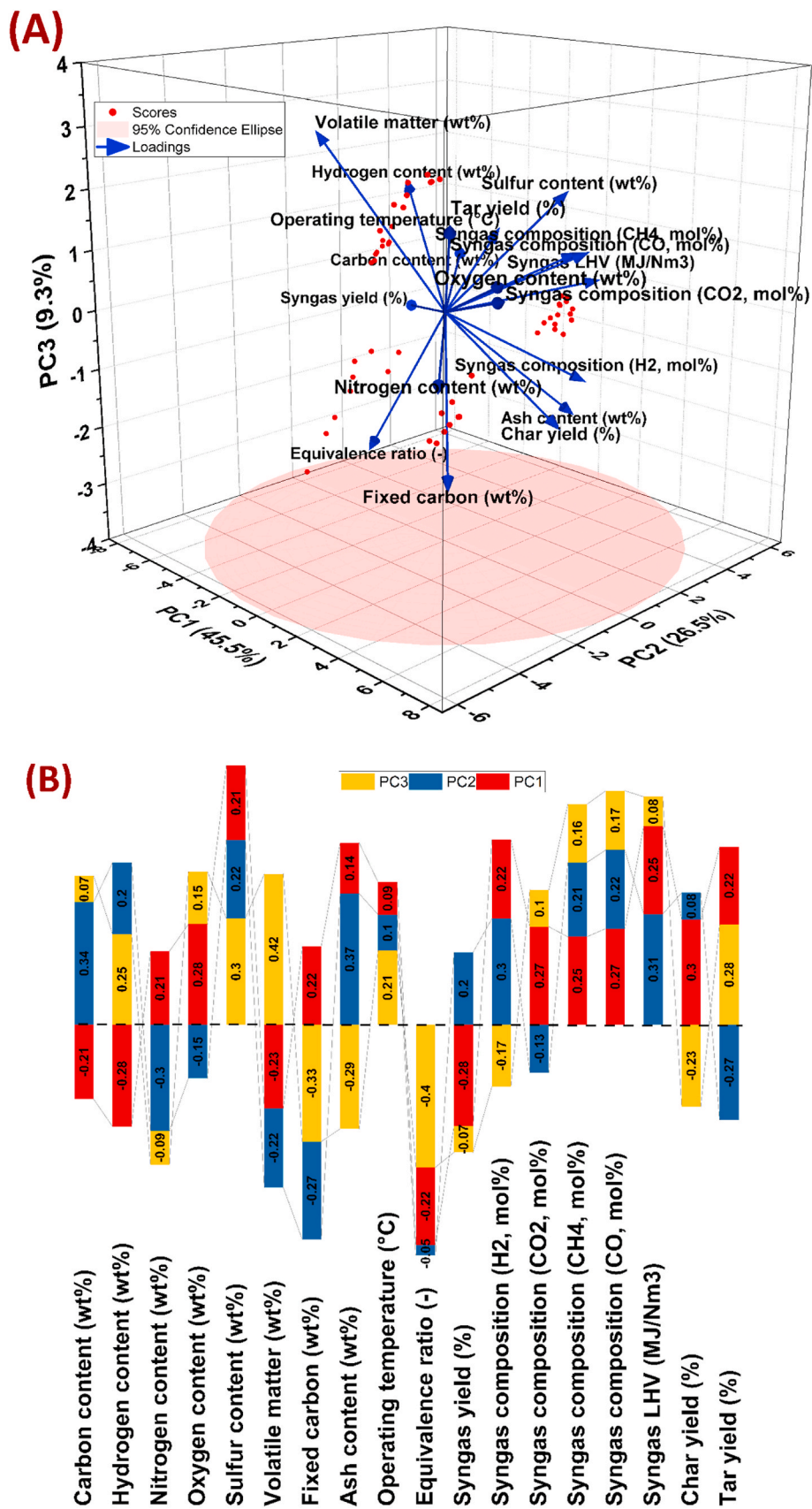


Fig. 3. (A) Principal component analysis of modeling data loading and score plots of input/output parameters, and (B) correlation of variables with the first three principal components.

26.86% and from 1.05 wt% to 46.70 wt%, respectively. The large variations in MSW composition can be attributed to a board diversity of MSW types. The operating temperature and equivalence ratio are in the range of 187–1412 °C and 0.1–1, respectively. The modeling data span a broad spectrum of possible gasification conditions. This comprehensive modeling data can provide an opportunity to develop a powerful ML approach with high generalization capability.

Syngas yield ranges between 21.5 and 92.8 wt%, while char and tar yields vary between 0.66–29.35 wt% and 0.4–55.35 wt%, respectively. The large variations in the product distribution in the MSW gasification process can be attributed to the large variations in feedstock characteristics and gasification parameters. CO contributes the most to the syngas composition, followed by CO₂, H₂, and CH₄. CO, CO₂, H₂, and CH₄ median values are 21.8, 16.35, 10.3, and 3.64 mol%, respectively. According to statistical analysis, the distributions of input variables and output responses are in reasonable ranges (neither concentrated nor dispersed), making them suitable for developing predictive models.

Spearman correlation analysis was conducted to identify the preliminary association between any two variables in the MSW gasification process. This analysis employs the rank values in modeling data to determine the strength and direction of the correlation between two parameters using a monotonic function [42]. Spearman correlation analysis is performed with an n-size feature set, and the ranking scores were represented for the X (first variable) and Y (second variable) columns. The highest number in each column receives a ranking of "1", while the lowest number receives the lowest ranking [26]. It should be noted that the positive and negative coefficient values revealed the influence trend of an input variable on a target variable. The value of −1 showed the perfect negative correlation, while the value of +1 indicated the perfect positive correlation in modeling data between two parameters. The Spearman correlation coefficient between selected columns (i.e., X and Y) was calculated according to the ranked values using the following equation:

$$r = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (1)$$

where r represents the Spearman's rank correlation coefficient, d_i denotes the difference between the two ranks of each observation, and n is the number of data.

Fig. 2 depicts the Spearman correlation matrix between input/output variables in the MSW gasification process. The operating temperature positively correlates with syngas yield ($r = 0.17$) since higher temperature promotes decomposition and cracking reactions [43]. It should be noted that increasing temperature favors H₂ and CO formation while reducing CO₂ and CH₄ generation due to the acceleration of Boudouard and water-gas shift reactions [44]. The equivalence ratio is determined by dividing the actual air-fuel ratio by the stoichiometric air-fuel ratio in the gasification process. This important parameter strongly affects the operating temperature, oxygen availability, and product distribution in the MSW gasification process. The equivalence ratio positively correlates with syngas yield ($r = 0.45$) since increasing this operating parameter increases the fuel oxidation rate and promotes feedstock conversion into syngas. Increasing the equivalence ratio has a beneficial effect on gaseous product formation by promoting the oxidation of volatile matter [45]. It should be noted that increasing the equivalence ratio enhances the energy efficiency of the gasification process while decreasing the LHV of resulting syngas [46]. On the contrary, a lower equivalence ratio value lowers the operating temperature in the gasification process. This can cause incomplete char-to-gas conversion, promoting char formation [47]. The equivalence ratio negatively correlates with H₂ and CO concentrations in syngas with a correlation coefficient of $r = -0.42$ and $r = -0.45$, respectively. It should be noted that lower H₂ and CO concentrations in the syngas can negatively reduce the calorific value of the resultant syngas.

The principal component analysis was conducted to unveil the

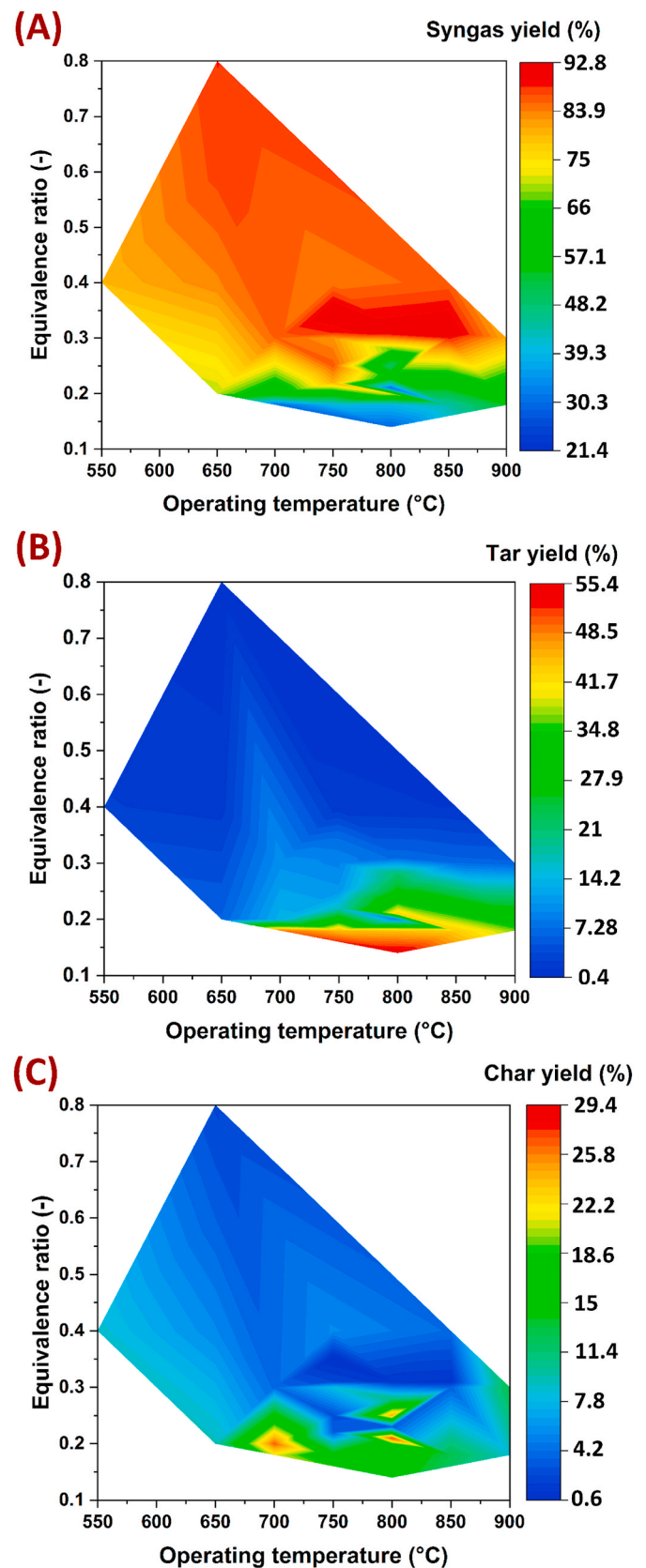


Fig. 4. Effects of the main operating parameters on product distribution during the MSW gasification process. (A) syngas yield (%); (B) tar yield (%); and (C) char yield (%).

Table 2

Statistical performance parameters of the developed ML models in modeling the MSW gasification process.

Training phase									
	Model type	Syngas yield (wt%)	Syngas composition H ₂ (mol%)	Syngas composition CO ₂ (mol%)	Syngas composition CH ₄ (mol%)	Syngas composition CO (mol%)	Syngas LHV (MJ/Nm ³)	Char yield (wt%)	Tar yield (wt%)
R ²	SVM	0.799	0.581	0.387	0.692	0.604	0.829	0.832	0.822
	RFR	0.947	0.950	0.930	0.956	0.970	0.930	0.931	0.959
	GBR	0.994	0.977	0.993	0.991	0.981	0.943	0.978	0.998
RMSE	SVM	18.35	7.68	14.31	3.500	13.332	1.088	5.169	14.195
	RFR	4.927	2.528	4.568	1.158	3.187	0.697	2.350	3.260
	GBR	3.131	2.441	2.811	0.926	3.509	0.709	1.778	2.431
RRMSE	SVM	0.276	0.541	0.649	0.653	0.539	0.172	0.492	0.831
	RFR	0.0716	0.180	0.211	0.229	0.128	0.112	0.228	0.212
	GBR	0.0457	0.177	0.125	0.172	0.149	0.114	0.165	0.141
Testing phase									
R ²	SVM	0.800	0.526	0.371	0.631	0.623	0.814	0.851	0.855
	RFR	0.934	0.936	0.906	0.950	0.961	0.922	0.925	0.955
	GBR	0.987	0.974	0.968	0.961	0.980	0.926	0.974	0.990
RMSE	SVM	16.742	7.736	15.139	3.172	15.735	1.183	7.558	14.596
	RFR	8.320	3.000	5.274	1.436	3.393	0.652	2.993	5.497
	GBR	6.318	2.243	3.397	1.216	4.459	1.060	2.888	3.995
RRMSE	SVM	0.231	0.568	0.710	0.791	0.714	0.188	0.555	0.954
	RFR	0.133	0.210	0.228	0.273	0.157	0.096	0.208	0.253
	GBR	0.099	0.147	0.169	0.304	0.165	0.159	0.229	0.267

relationships among all parameters in the MSW gasification process. In this analysis, features with similar scores would be placed in the same direction (i.e., positive importance), while descriptors with dissimilar scores would be placed in the opposite direction (i.e., negative importance). Principal component analysis results for modeling data are shown in Fig. 3. The position of operating temperature and syngas yield in the loading plot is relatively close, indicating a positive correlation between them (Fig. 3A). In general, the decomposition and cracking of MSW into gaseous products are significantly higher at higher operating temperatures during the gasification process. The first and second principal components account for about 45.5% and 26.5% of the total variance, respectively. In addition, the cumulative percentage of the variance of the first three principal components exceeds 81.3%. This result demonstrates the ability of the first three principal components to adequately reflect the vast majority of information in the original data. It is worth noting that the first principal component mainly comprises factors such as the hydrogen and oxygen content of MSW, whereas the second principal component consists primarily of fixed carbon and ash content (Fig. 3B). These two principal factors emphasize the importance of ultimate and proximate analyses in the first and second principal components, respectively.

Char yield positively correlates with ash content because of a considerable amount of inorganic materials in ash-rich biomass feedstocks. These inorganic materials, such as alkali and alkali-earth metals, can not be converted to gaseous/liquid products while remaining in solid residue. The ash content also significantly impacts the gasification process since the higher ash content causes serious operational problems like passage fouling, slagging, and blocking [48]. The oxygen content is positively correlated with tar yield because tar components contain a wide spectrum of aromatics, ranging from monocyclic to bicyclic, and other oxygen-containing species and complex polycyclic aromatics [49]. It should be noted that these correlations might change based on MSW characteristics and gasification parameters.

Fig. 4 depicts the contour diagram illustrating the relationship between the main operating factors in the MSW gasification process (i.e., reaction temperature and equivalence ratio) and product distribution at different phases. The operating temperature and equivalence ratio in the range of 730–800 °C and 0.3–0.4, respectively, result in higher syngas yields. According to Le Chatelier's principle, increasing the reaction temperature can enhance the products of endothermic reactions. More specifically, higher gasification temperature can promote endothermic reactions, such as boudouard, water–gas shift, and steam reforming.

Furthermore, char and tar concentrations are decreased at higher gasification temperatures, while syngas yield is increased because of an increase in the decomposition rate of feedstock compositions and secondary cracking reactions.

3.2. Modeling results

The statistical performance parameters (R², RMSE, and RRMSE) of the developed ML models are reported in Table 2. The GBR model shows the best performance in training and testing because of having the highest R² values and the lowest RMSE and RRMSE values. More specifically, the GBR model can model the MSW gasification process better than the RFR and SVM models, with R² = 0.926–0.990, RMSE = 1.06–6.318, and RRMSE = 0.099–0.304 in the testing phase. The SVM model is the weakest approach, with R² = 0.371–0.855, RMSE = 1.188–16.742, and RRMSE = 0.188–0.954 in the testing phase. Generally, the GBR technique can satisfactorily model all the outputs of the MSW gasification process because of lower errors and the coefficient of determination values close to unity. A detailed look at the obtained statistical parameters indicates that the coefficient of determination values in the testing phase are lower than those in the training phase for each specific output. So, it can be inferred that there is no overfitting in the GBR modeling process.

The fidelity of the developed ML models is also evaluated by displaying the estimated values against the actual ones for both the training and testing steps (Fig. 5). In Fig. 5, the blue dots and lines refer to the training phase, and the red ones belong to the testing step. Both red and blue lines have a colorful shadow, representing a 95% confidence level of the prediction interval. The dashed line represents a 45-degree line. Data points are distributed closely enough around a 45-degree line for all the outputs for both the GBR and RFR models, but a better distribution can be seen in the GBR model. In addition, The GBR-predicted values fit better with experimental data with an observable linear association (Fig. 5).

Overall, the GBR model shows the best prediction performance, followed by the RFR model. However, the widely used SVM model has the weakest prediction performance and accuracy compared to the other models. Additionally, the GBR model can provide better accuracy and reliability than the previously published paper because of the consistency and homogeneity of the collected data patterns [23]. Another reason can be the complexity and high nonlinearity of MSW gasification modeling. By contrast, the GBR model can accurately predict the

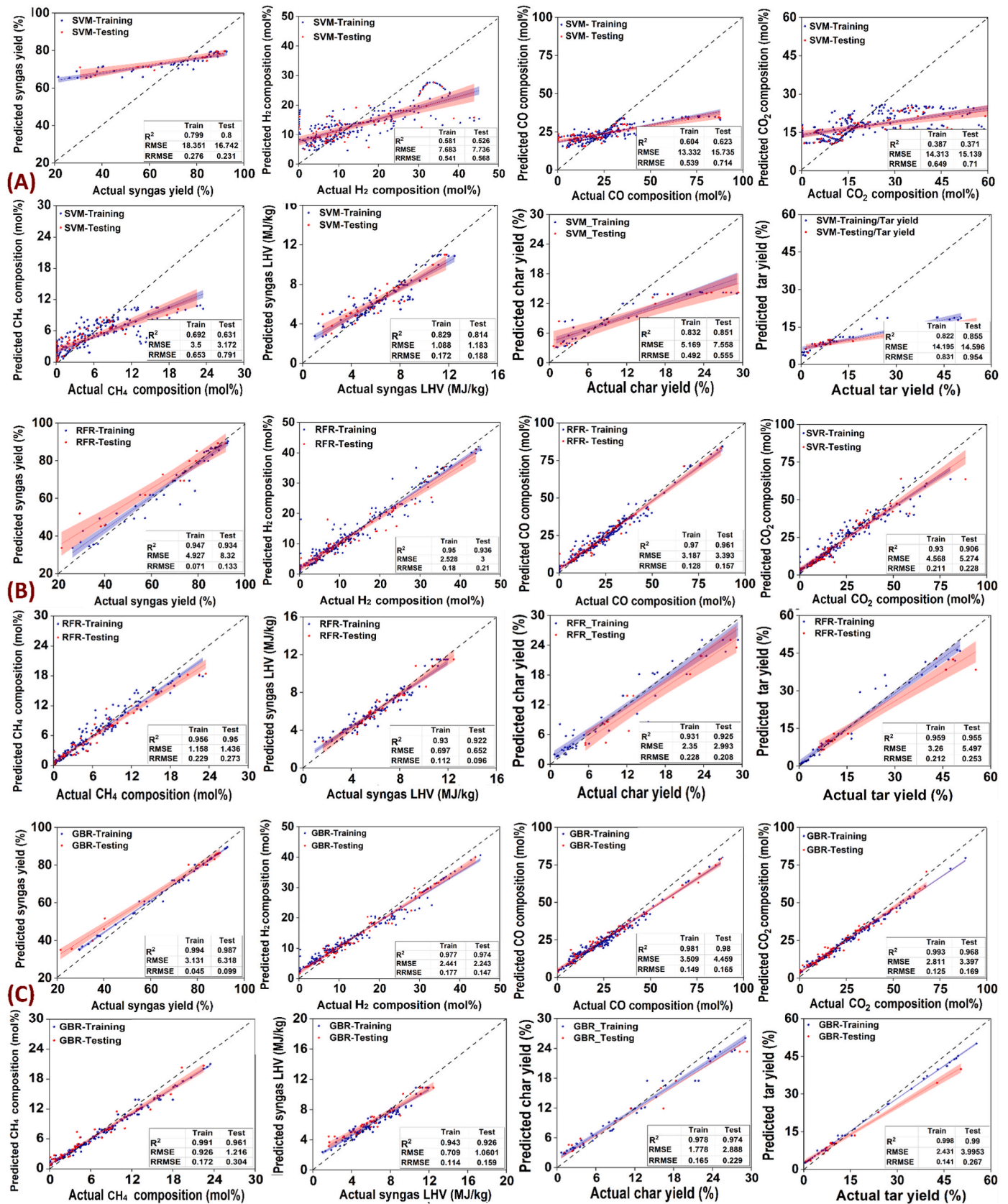


Fig. 5. Predicted data against experimental values during training and testing of (A) SVM, (B) RFR, and (C) GBR models.

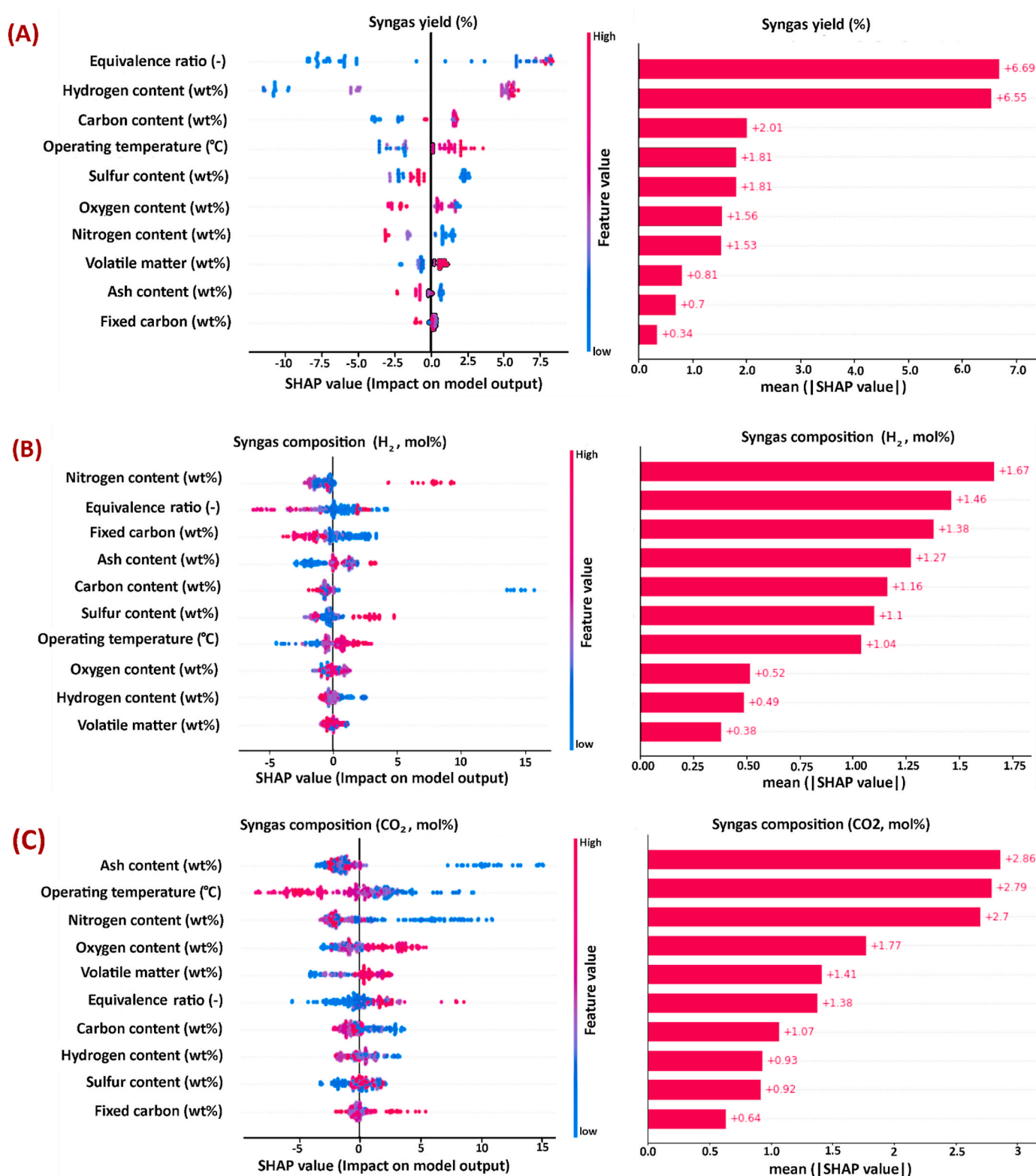


Fig. 6. SHAP value indicating the impacts of input parameters on each individual output during the MSW gasification process. (A) syngas yield; (B) syngas composition (H_2); (C) syngas composition (CO_2); (D) syngas composition (CH_4); (E) syngas composition (CO); (F) syngas lower heating value (LHV), (G) char yield; and (H) tar yield.

nonlinearity and complexity of the process. The obtained results can be due to the nature of GBR with randomly selecting trees during the training while lowering the probability of overfitting and improving the accuracy. In addition, compiling a decision tree at each step during the

prediction can make the model more reliable and robust than other classic decision tree regressors [50]. The GBR model can also incorporate automatic variable selection and model choice in the fitting process. This ML model can be more flexible and stable than most conventional

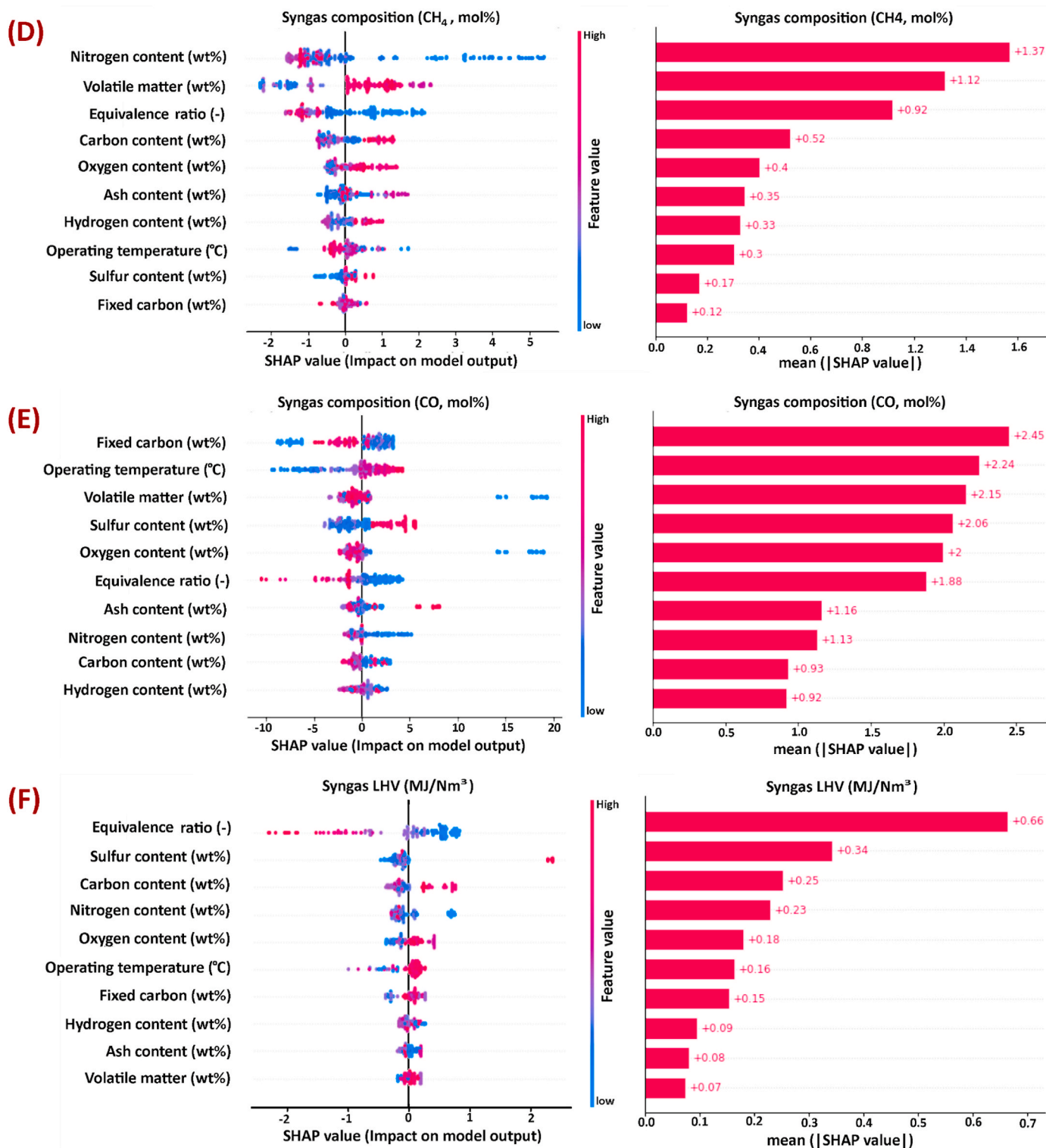


Fig. 6. (continued).

prediction algorithms when faced with a high-dimensional dataset [51].

3.3. SHAP analysis

The black-box nature of ML models makes it difficult to understand the significance and contribution of input descriptors on target outputs in the modeling process. To cope with this issue, SHAP analysis was performed using the best-performing model (*i.e.*, the GBR algorithm) to

determine the importance of input parameters and their optimal interval for each predicted target. This powerful analysis is an unbiased and consistent method to have meaningful insights into the outcomes of ML models from both local and global points of view [52]. The fundamental principle of the SHAP analysis is that, for each prediction, an importance value for each feature variable is calculated, providing local interpretability. Furthermore, the overall importance of input features can be realized by adding up the absolute SHAP values across the entire dataset,

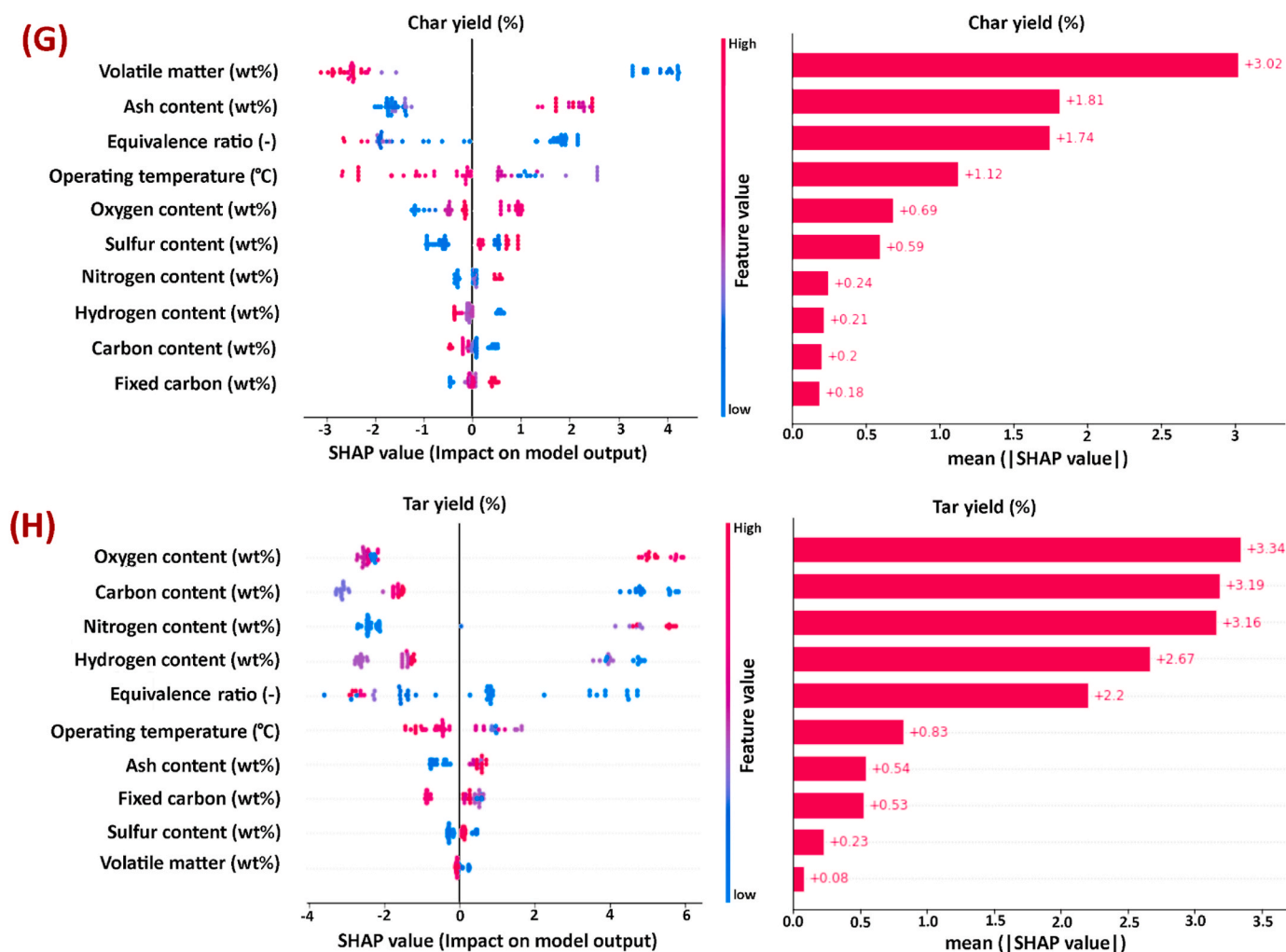


Fig. 6. (continued).

providing global interpretability [53]. This promising approach has a strong theoretical foundation in cooperative game theory and aids in investigating nonlinear interactions of the features in complex processes.

The distribution of SHAP values for each feature during the MSW gasification process is illustrated in Fig. 6. The order of input descriptors is defined as the mean absolute SHAP values imposed on predicted responses, with higher SHAP values ranking first and *vice versa*. Positive SHAP values show that the input features contribute positively to the predicted values, whereas negative SHAP values suggest that the corresponding input predictors are affected negatively. In the left-hand graphs, the red-colored points indicate a high value of the input descriptor, while the blue-colored show a low value. Notably, changing data points from red to blue by increasing the SHAP value indicates that the input descriptor has a negative influence on the output target and *vice versa*. The right-hand bar charts represent the global importance of input variables on each individual output across the dataset based on the mean absolute SHAP value.

According to modeling and SHAP analysis, the most important influencing factors for syngas production are the carbon and hydrogen content of MSW (compositional parameters) and the equivalence ratio and temperature of the process (operating parameters). Volatile matter and ash content of MSW, along with the equivalence ratio and temperature of the process, are the most important parameters for char formation. Nevertheless, the oxygen, carbon, nitrogen, and hydrogen content of MSW are the top four parameters affecting tar yield. As a

result, selecting appropriate feedstocks and operating conditions is important for optimizing the MSW gasification process.

As shown in Fig. 6A, the equivalence ratio of the process and the hydrogen content of feedstock are the top two most significant parameters on syngas production during gasification, with an average SHAP value of 6.69 and 6.55, respectively. Therefore, a higher equivalence ratio results in a higher positive SHAP value, implying that it favorably affects syngas yield. Increasing the equivalence ratio of the process increases the likelihood of oxygen reacting with volatiles to form gaseous products during gasification. The equivalence ratio of the process should be carefully optimized since its too-low values result in lower reaction temperature, and its too-high values consume more combustible gases through oxidation reaction, decreasing the efficiency and productivity of the process (Fig. 6F) [54]. It should be noted that at higher equivalence ratios, the concentration of H_2 and CO in the syngas is significantly lowered, while CO_2 content is increased because of an increase in the rate of oxidation reactions (see Fig. 6B–E) [47]. The fixed carbon content shows no obvious monotonicity with CO_2 and CH_4 concentrations, and either a low or high fixed carbon content can be associated with higher CO_2 and CH_4 concentrations.

Fig. 6(G–H) illustrates the SHAP values of input parameters on the yield of char and tar during the MSW gasification process. The volatile matter and ash content are the top two impactful input features, where the volatile matter has a negative impact, and ash content positively affects the char yield. The higher inorganic compounds in ash-rich feedstocks can increase char yield, decrease char stability/porosity,

and increase agglomeration during the MSW gasification process [55, 56]. The feedstocks with higher ash contents improve the tar conversion while contaminating the syngas with more mineral dust [57]. The oxygen, carbon, and nitrogen contents of MSW emerge as the three most important input features in tar formation. More specifically, the oxygen content of MSW is positively correlated with tar yield, while carbon content exhibits a non-monotonicity.

4. Conclusions

In this study, the MSW gasification process was modeled and scrutinized using ML technology. To this end, an inclusive dataset containing 317 data patterns (including MSW characteristics and gasification parameters) was obtained from the eligible published papers. The compiled dataset was thoroughly analyzed using statistical methods to show an overall picture of the effect of the input descriptors on the output targets. The collected data patterns were then subjected to three ML models (i.e., SVR, RFR, and GBR) to estimate product distribution (syngas, char, tar yield) and syngas quality as a function of biomass characteristics and gasification parameters. The highest prediction accuracy ($R^2 > 0.926$, RMSE < 6.318 , and RRMSE < 0.304) was achieved when the GBR model was used in the modeling process. The significance and contribution of input descriptors on target outputs in the modeling process were successfully understood using the SHAP analysis. The equivalence ratio of the process and the hydrogen content of MSW contributed the most to the prediction of syngas yield and quality in the GBR model. Overall, ML technology could reduce experimental costs and efforts in finding optimal biomass characteristics and process conditions in MSW gasification.

MSW gasification systems can be evaluated for their economic feasibility and environmental impact using ML models developed based on experimental data. The developed models could also pave the way for real-time inspection and control of industrial MSW gasifiers. Despite the promising results obtained herein, developing more comprehensive models to predict MSW gasification is still possible. This study did not consider the ash components capable of acting as catalysts for reactions. The accuracy and reliability of MSW gasification models can be improved with more detailed information about the ash content. Models developed herein are applicable only to non-catalytic MSW gasification. Therefore, it is necessary to develop models applicable to MSW gasification both catalytically and non-catalytically in future investigations.

Credit author statement

Yadong Yang: Investigation, Formal analysis, Writing - Original Draft. Hossein Shahbeik: Investigation, Writing - Original Draft. Alireza Shafizadeh: Validation, Visualization. Shahin Rafiee: Software, Methodology. Amir Hafezi: Data Curation, Methodology. Xinyi Du: Software, Methodology. Junting Pan: Funding acquisition, Supervision, Project administration, Writing - Review & Editing. Meisam Tabatabaei: Conceptualization, Supervision, Resources, Project administration, Writing - Review & Editing. Mortaza Aghbashlo: Conceptualization, Supervision, Resources, Project administration, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2023.127881>.

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